Method protocol No 5

Method name: Level of free radical generation (ROS) version: 0

1. Materials needed

Cell line: EA.hy 926 (**ATCC® CRL 2922)**

- Growth Medium: Dulbecco's Modified Eagle's Medium ATCC Cat. No 30-2002
- (4 mM L-glutamine, 4500 mg/L glucose, 1 mM sodium pyruvate, and 1500 mg/L sodium bicarbonate), Biowest Cat.No L0104, Corning Cat No 10-013-CV
- Fetal Bovine Serum Heat Inactivated, Biowest Cat No S181H
- Dulbecco's Phosphate Buffered Saline (DPBS), Biowest Cat. No L0615
- Trypsin 0,25%, Biowest Cat. No L0931 (w/o Calcium, w/o Magnesium, w/o Phenol Red, Sterille, Filtered)
- Penicilin-Sretpomycin-Neomycin Solution, SigmaAldrich Cat. No P4-83 (~5,000u penicillin, 5ng streptomycin, 10mg neomycin/ml)
- Trypan blue (0,4%)
- 6-carboxy-2',7'-dichlorodihydrofluorescein diacetate, Invitrogen Cat. No C400
- 100µM solution H_2O_2, POCh Poland

2. Needed devices and tools

- laminar flow cabinet
- microplate reader Victor
- CO_2 incubator
- laboratory centrifuge
- water bath
- automatic cell counter
- automatic pipettes
- sterile tips for automatic pipettes
- serological pipettes
- sterile Pasteur pipettes
- 75cm^2 culture dishes
- pipettor
- gas burner
- cell counting slides
- 96-well tissue culture plates
- 37°C water bath

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3. Preparation of sample for testing

   Sterile samples of nanomaterials (The sterilization method will be provided later)

4. Experimental conditions

   Cells incubation - 37°C in a 5% CO_2 in humid air atmosphere

5. Course of the experiment

Prepare an appropriate number of flasks with EA.hy-926 cells. The cell confluence should reach about 80-90%.

1. Seed cells in 96-well tissue culture plates. Use a complete culture medium. Density per well 10^4 cells/well. Five wells should be allocated for each sample understood as one specific concentration of a given nanomaterial. In addition, a negative control (without addition of test nanomaterial, 5 wells).
2. Incubate cells in incubator for 24 h.
3. Replace the complete culture medium with 100µl serum-free medium.
4. Add suspension of nanoparticles under test at a concentration of .... /well.
   (the amount of nanoparticles will be determined in the course of the experiment).
5. Incubate cells with nanoparticels in incubator in standard conditions for 24 h.
6. Add 10 µl of 100µM solution H_2O_2 to each well of the positive control.
7.
8. Prepare H_2DCFDA (carboxyfluorescin diacetate) 100 µM stock solutions in ethanol.
9. Add 10 µl of H_2DCFDA stock solution to each well (final a concentration of 10 µM) and followed by 30 min incubation at 4˚C in the dark before scanning.
10. Fluorescence reading at 485/530nm.

6. Source data format and access path

   Will be provided

7. Analysis and interpretation of results (software used)

   Will be provided as experiments progress

8. Comments and remarks

   Will be provided as experiments progress

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